

Basics of Databases

Parteek Kumar

DBMS



Understanding Data and Information

Data	Information
Data is raw facts and figures	Processed form of data
23 is data	Age 23 is information
Data cannot be used for decision making	Information is useful for decision making
Data is atomic.	Information is collection of data.

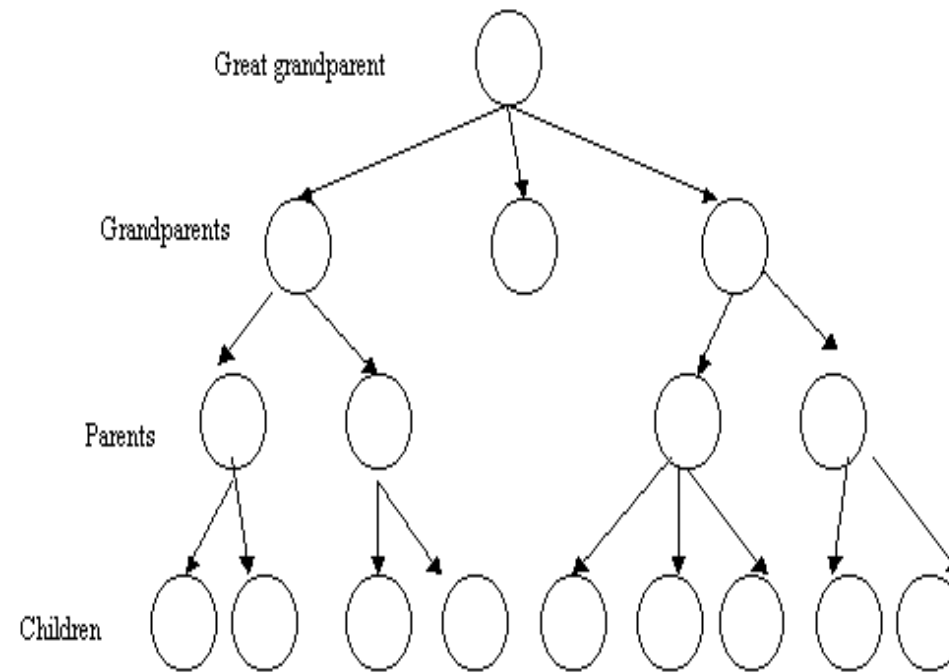
Database

The related information when placed in an organized form makes a database.

Ways to Organize the Information

Conversion of Data to Information

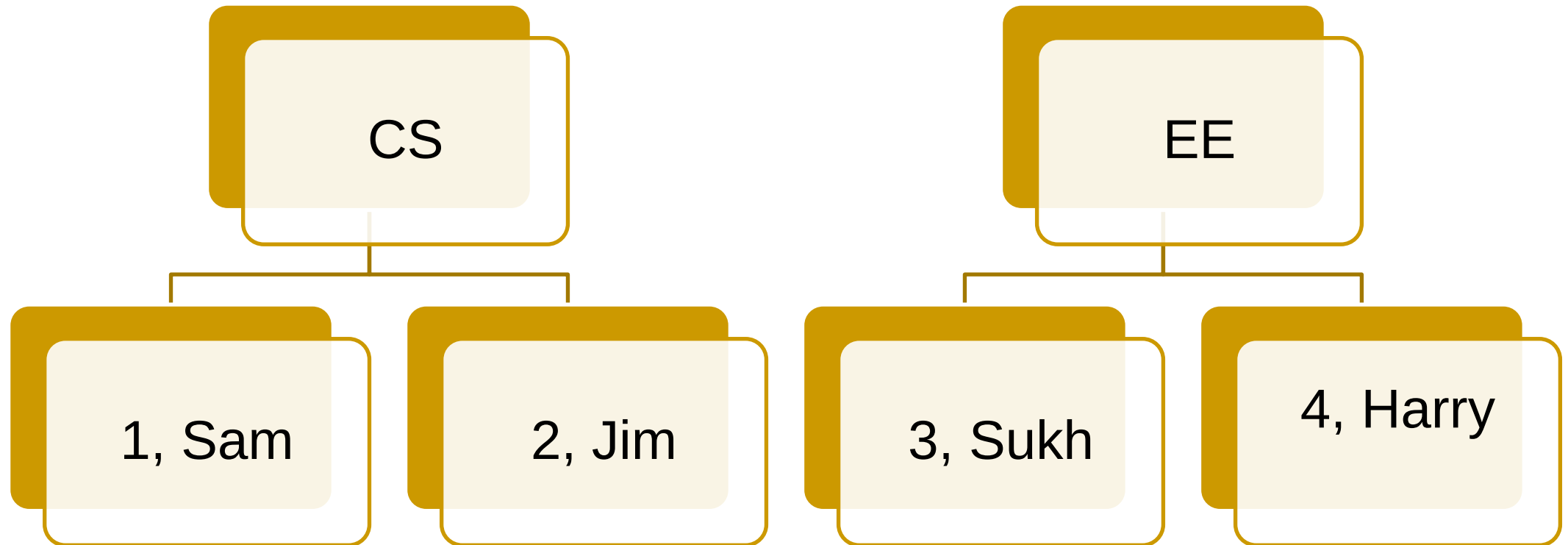
■ Tree Data Structure



Ways to Organize the Information

Conversion of Data to Information

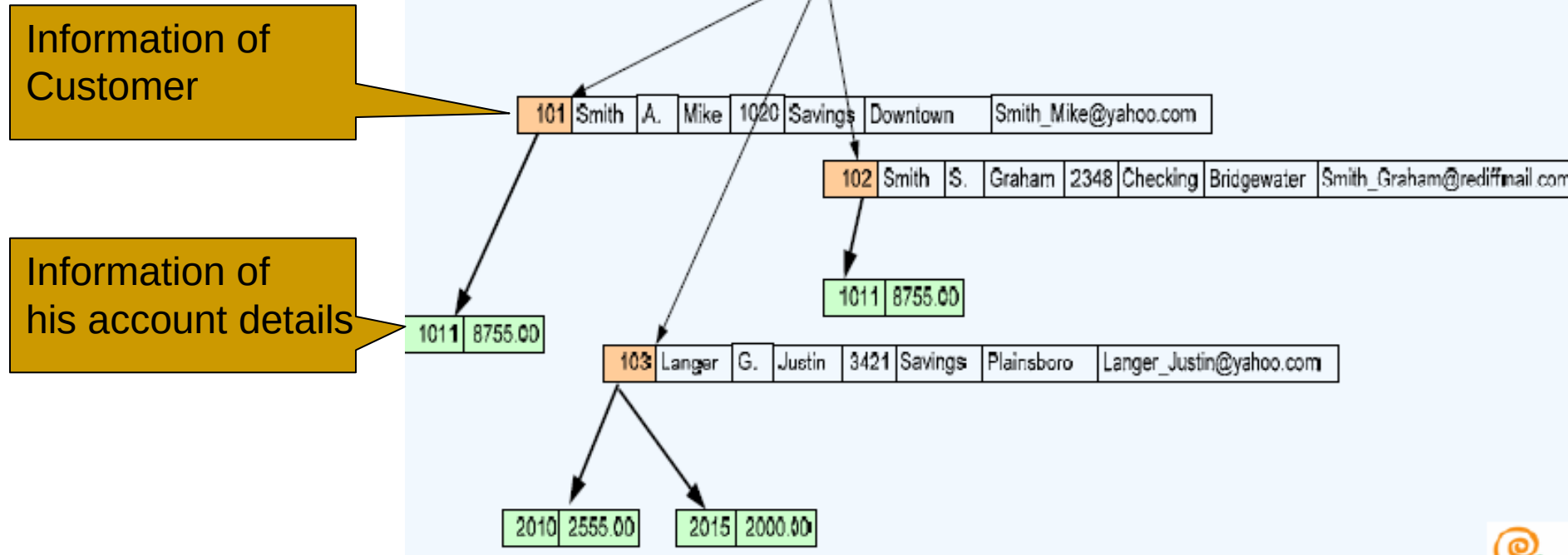
- Tree Data Structure



Ways to Organize the Information

Conversion of Data to Information

- Tree Data Structure: Information of Customers and their Account details



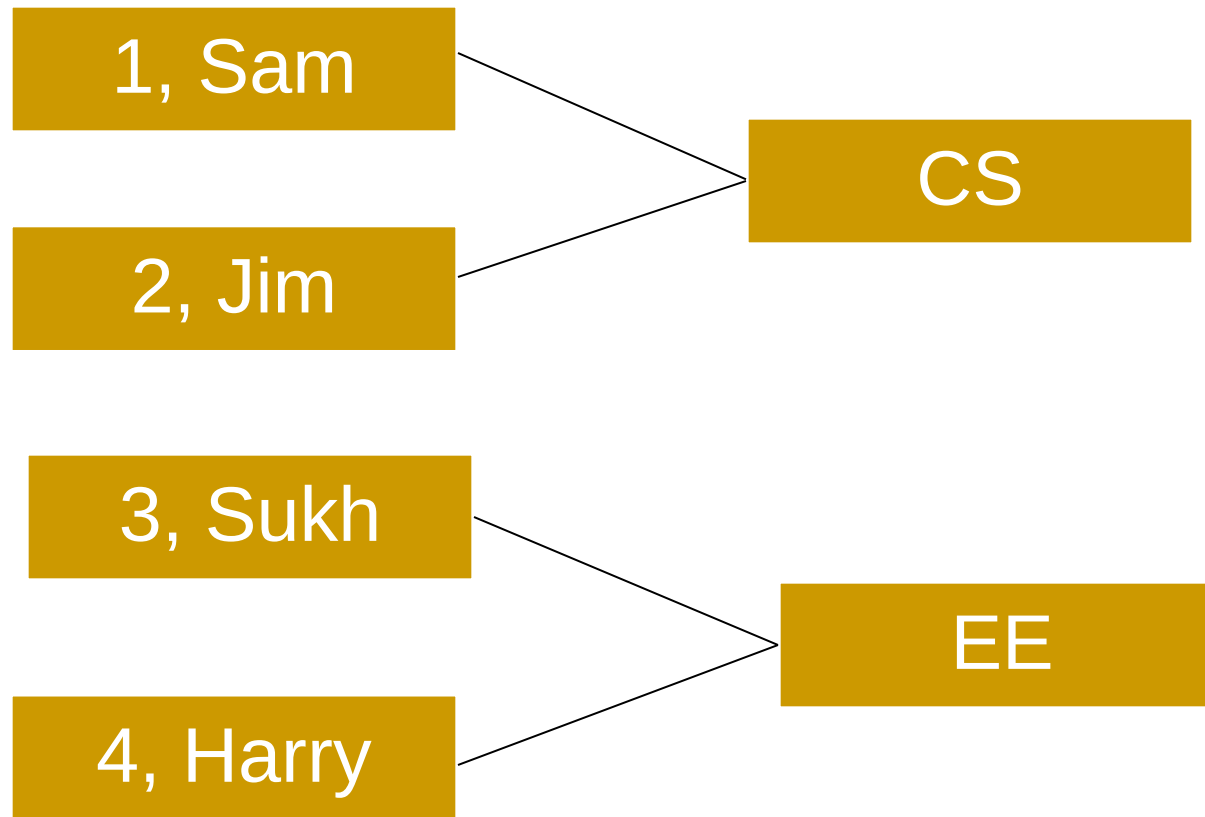
Ways to Convert Data to Information

- Data Structure
 - Tree
- Data Model
 - Hierarchical Model

Ways to Organize the Information

Conversion of Data to Information

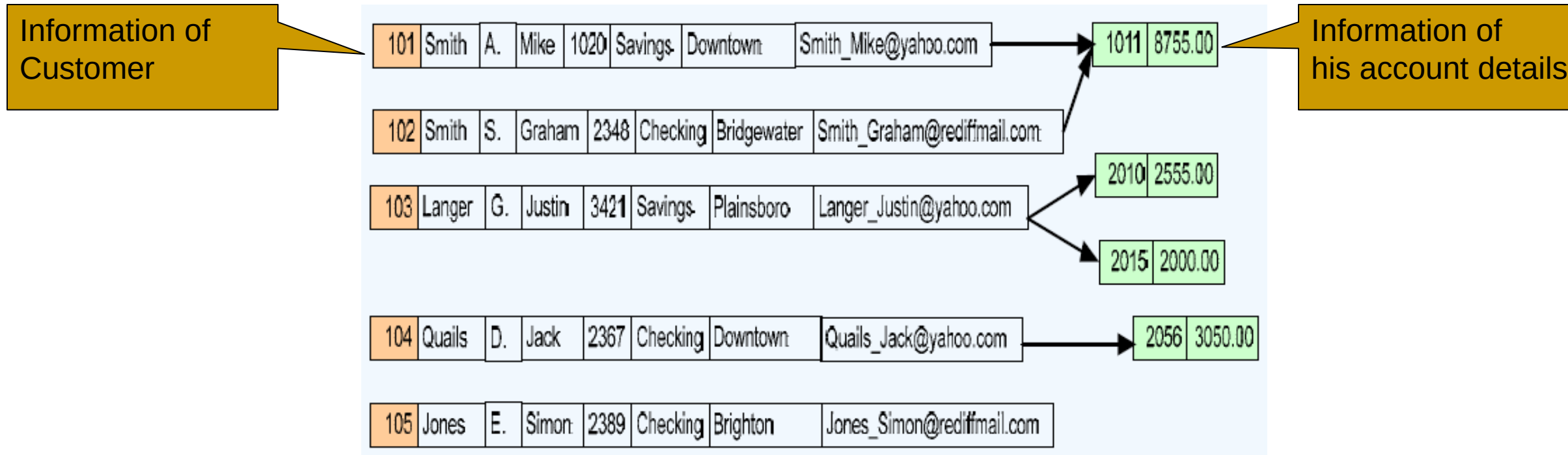
- Graph Data Structure



Ways to Organize the Information

Conversion of Data to Information

- Graph Data Structure: Information of Customers and their Account details



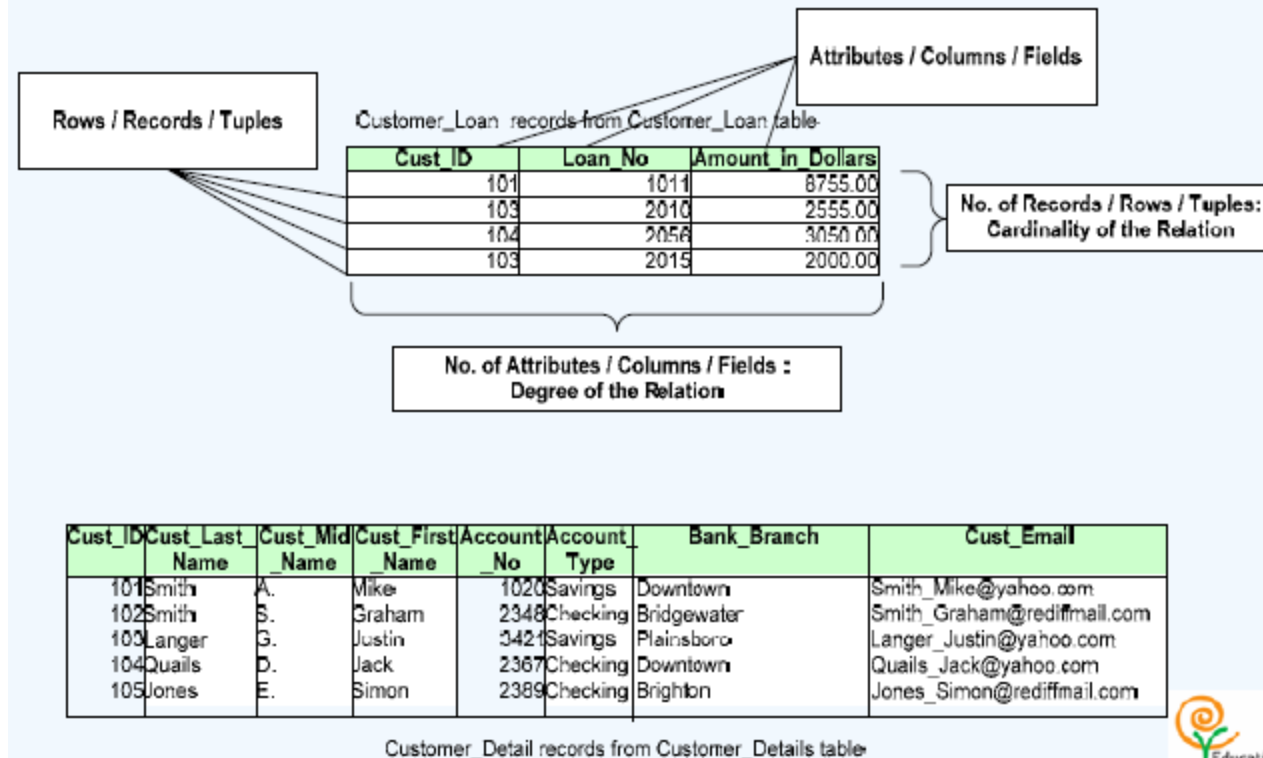
Ways to Convert Data to Information

- Data Structure
 - Graph
- Model
 - Network Model

Ways to Organize the Information

Conversion of Data to Information

- Arrays Data Structure: Information of Customers and their Account details



Ways to Organize the Information

Conversion of Data to Information

■ Arrays Data Structure

Rno	Name	Class
1	Rahat	BE(CSE)
2	Rishan	BE(CSE)
3	Sukh	BE(ECE)
4	Harman	BE(ECE)

Ways to Organize the Information

Conversion of Data to Information

- Data Structure
 - Array/Table
 - Model
 - Relational Model
-

Ways to Organize the Information

Conversion of Data to Information

■ Three Data Models

- ❑ Hierarchical Data Model: Based on tree data structure
- ❑ Network Data Model: Based on graph data structure
- ❑ Relational Data Model: Based on Arrays/tables

Relational Data Model is simple and easy to use.

Thus, out of all data models Relational Data model is commonly used. And popular software like Oracle, MySQL, MS Access are based on this model.

DBMS: Management

- It is management of information.
- We can perform Insert, Update, Delete and Retrieve operations over database.

Operations on Databases

- To add new information (e.g. to add the address of a new friend in your address book)
- To view or retrieve the stored information (e.g. you have to find the address of one of your old friends)
- To modify or edit the existing information (e.g. your friend has shifted to a new place so his address would get changed)
- To remove or delete the unwanted information (e.g. your friend has changed his/her mobile number, so his/her mobile number would have to be removed from list)
- Arranging the information in a desired order etc.

DBMS: System

- A system or software which manage the database on a computer.

DBMS

- A software responsible to manage database, i.e., a software responsible to manage insert, update, delete and retrieve operations over database in a computer is known as Database Management System or DBMS.
- Examples: Oracle, MySQL, Sybase, SQL Server, FoxPro etc.

File Based System Vs. Centralized Database System and Advantages and Disadvantages of DBMS

Maintaining Database on Computers

- There are two approaches
 - Traditional File Based Approach
 - Centralized DBMS approach

File Based Approach

- It is an early attempt to computerize the manual filing system. It is first implemented by using file handling utilities of COBOL (Common Business Oriented Language).
- Data is stored in flat files. For example: Text files, csv files
- Each file called a flat file, contained and processed information for one specific function.
- Now a days file Handling utilities of Programming languages like C, C++, Java etc. is used to implement it.

File Based Approach

Ways of storing data in files – customer data

4176	Aniruddha Sarkar	SBU1
4181	Manoj Saha	SBU1
4183	Moushumi Dharchoudhury	SBU1
4203	Suryanarayana D.V.S.S.	SBU1
4204	Vivek Rai	SBU1

Predefined

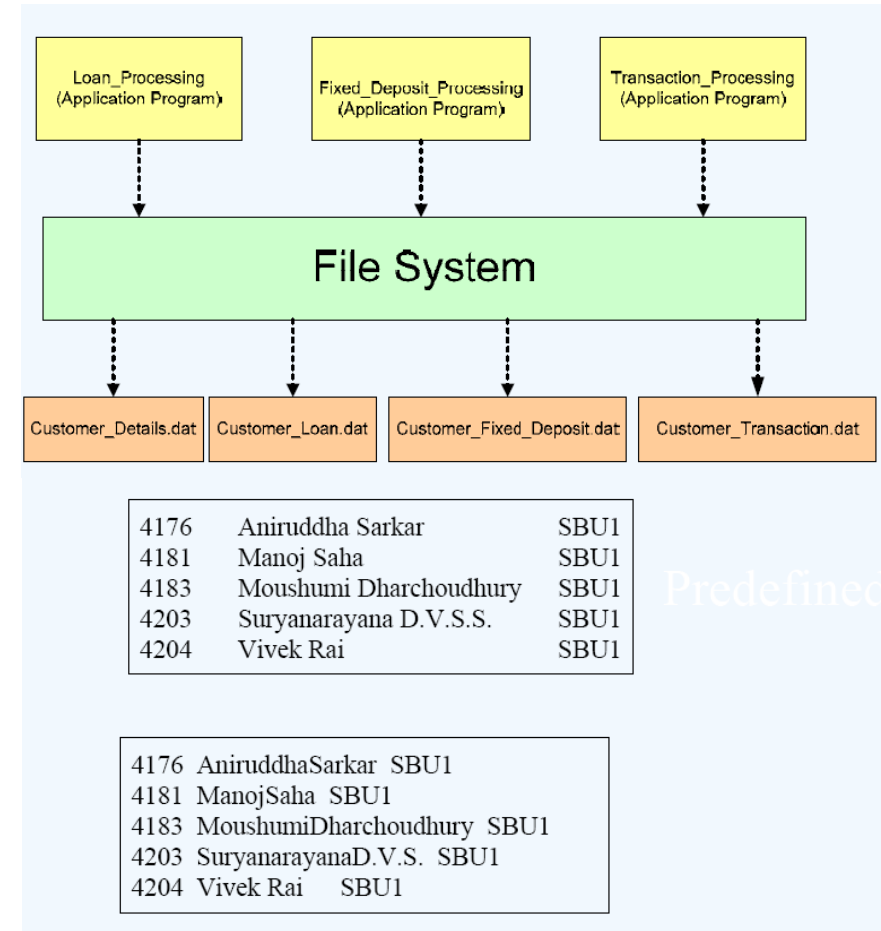
4176	AniruddhaSarkar	SBU1
4181	ManojSaha	SBU1
4183	MoushumiDharchoudhury	SBU1
4203	SuryanarayanaD.V.S.	SBU1
4204	Vivek Rai	SBU1

Limitations of the File-Based Approach

- **Separated and Isolated Data**
 - **Duplication of data**
 - **Data Dependence**
 - **Difficulty in representing data from the user's view**
 - **Data Inflexibility**
 - **Incompatible file formats**
-

Limitations of the File-Based Approach

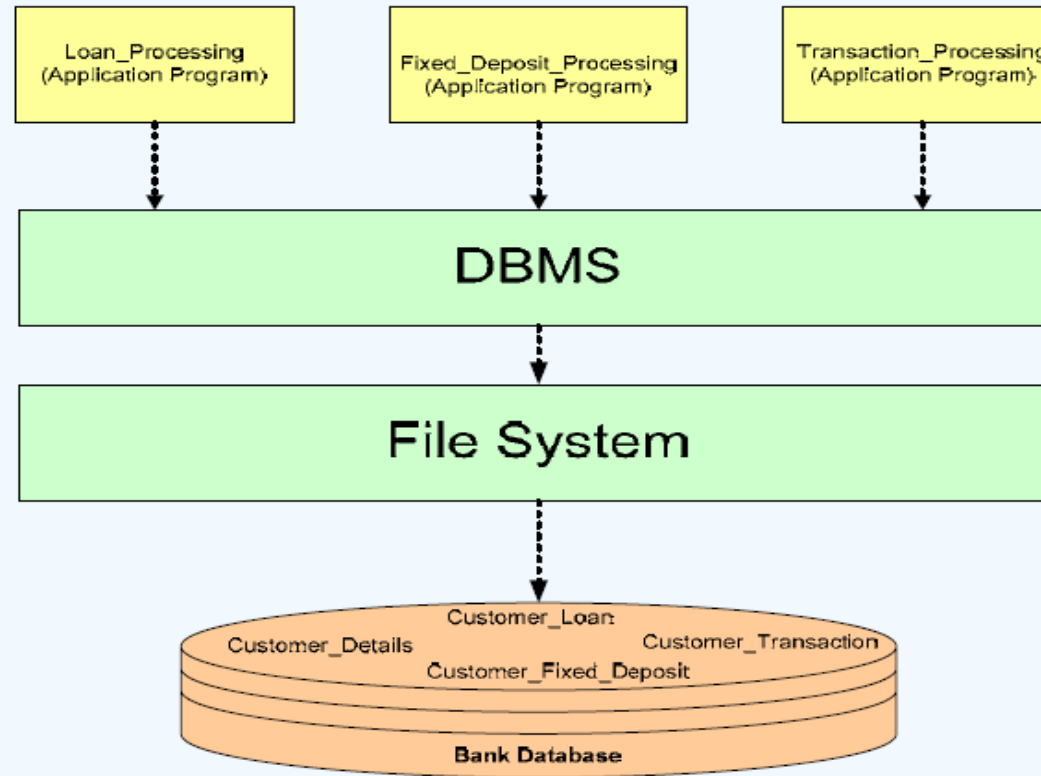
- Separated and Isolated Data
- Duplication of data
- Data Dependence
- Difficulty in representing data from the user's view
- Data Inflexibility
- Incompatible file formats



Database Approach

Where does the DBMS fit in?

Position of DBMS



Example: File Based System Class Activity

Identify Issues in this database

General Office	Library	Hostel	Account Office
Rollno Name Class Father_Name Date_of_birth Address Phone_No Previous_Record Attendance Marks etc.	Rollno Name Class Address Date_of_birth Phone_No No_of_books_issued Fine etc.	Rollno Name Class Father_Name Date_of_birth Address Phone_No Mess_Bill RoomNo etc.	Rollno Name Class Address Phone_No Fee Installments Discount Balance Total etc.

Advantages of DBMS

- **Controlling Redundancy**

Example: Database Approach: Class Activity

How earlier identified issues are handled in this?

General Office	Library	Hostel	Account Office
Rollno Name Class Father_Name Address Phone_No Date_of_birth Previous_Record Attendance Marks etc.	Rollno No_of_books_issued Fine etc.	Rollno RoomNo Mess_Bill etc.	Rollno Fee Installments Discount Balance Total etc.

Integrity can be enforced

- Integrity of data means that data in database is always accurate, such that incorrect information cannot be stored in database.

Inconsistency can be avoided

- Data of same entity should be same at all the places.
- Duplication of data may results in to inconsistency as the two entries regarding the same data may not agree. At such times the data is said to be inconsistent.

Other Advantages

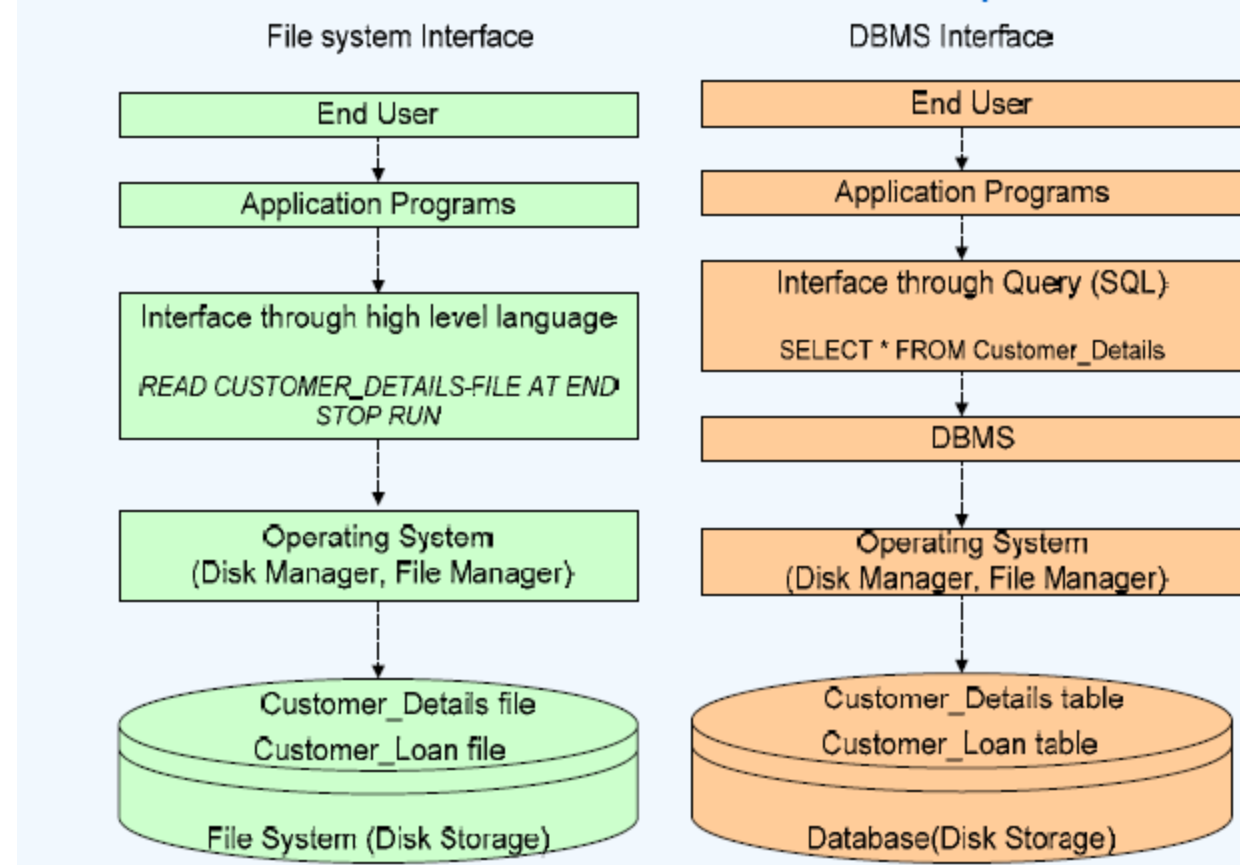
- Data can be shared
 - Providing Backup and Recovery
 - Standards can be enforced
 - Restricting unauthorized access
 - Solving enterprise requirement than individual requirement
-

Disadvantages of DBMS

- Complexity
 - Size
 - Performance
 - Higher impact of a failure
 - Cost of DBMS
 - Additional Hardware costs
 - Cost of Conversion
-

Comparison of File Management System with Database Management System

Difference Between File and DBMS Operations



Comparison of File Management System with Database Management System

File Management e.g. C++ or COBOL program	Database Management e.g. Oracle or Sybase
Small systems Relatively cheap Few 'files' Files are files Simple structure Redundant data Chances of inconsistency Isolated data Little preliminary design Integrity left to application programmer No security Simple, primitive backup/recovery Often single user	Large systems Relatively expensive Many 'files' Files are tables Complex structure Reduced redundancy Consistent Data can be shared Vast preliminary design Rigorous inbuilt integrity checking Rigorous security Complex & sophisticated backup/recovery Multiple users

When not to Use a DBMS

The overhead costs of using a DBMS are due to the following:

- High initial investment in hardware, software, and training
 - Overhead for providing security, concurrency control, recovery, and integrity functions.
 - Additional problems may arise if the database designers and DBA do not properly design the database or if the database systems applications are not implemented properly.
-



Components of DBMS

Components of DBMS

- Hardware
- Software
- Data
- Users
- Procedures

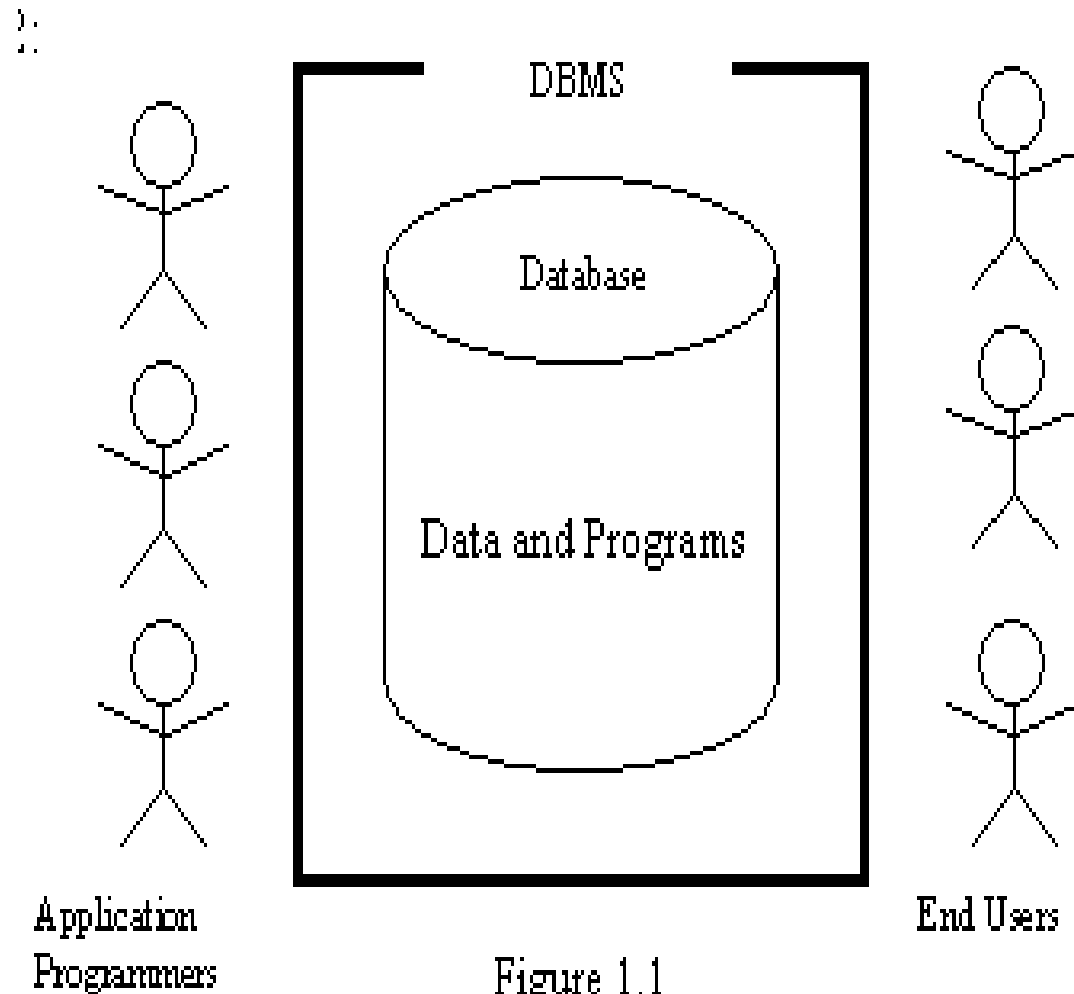
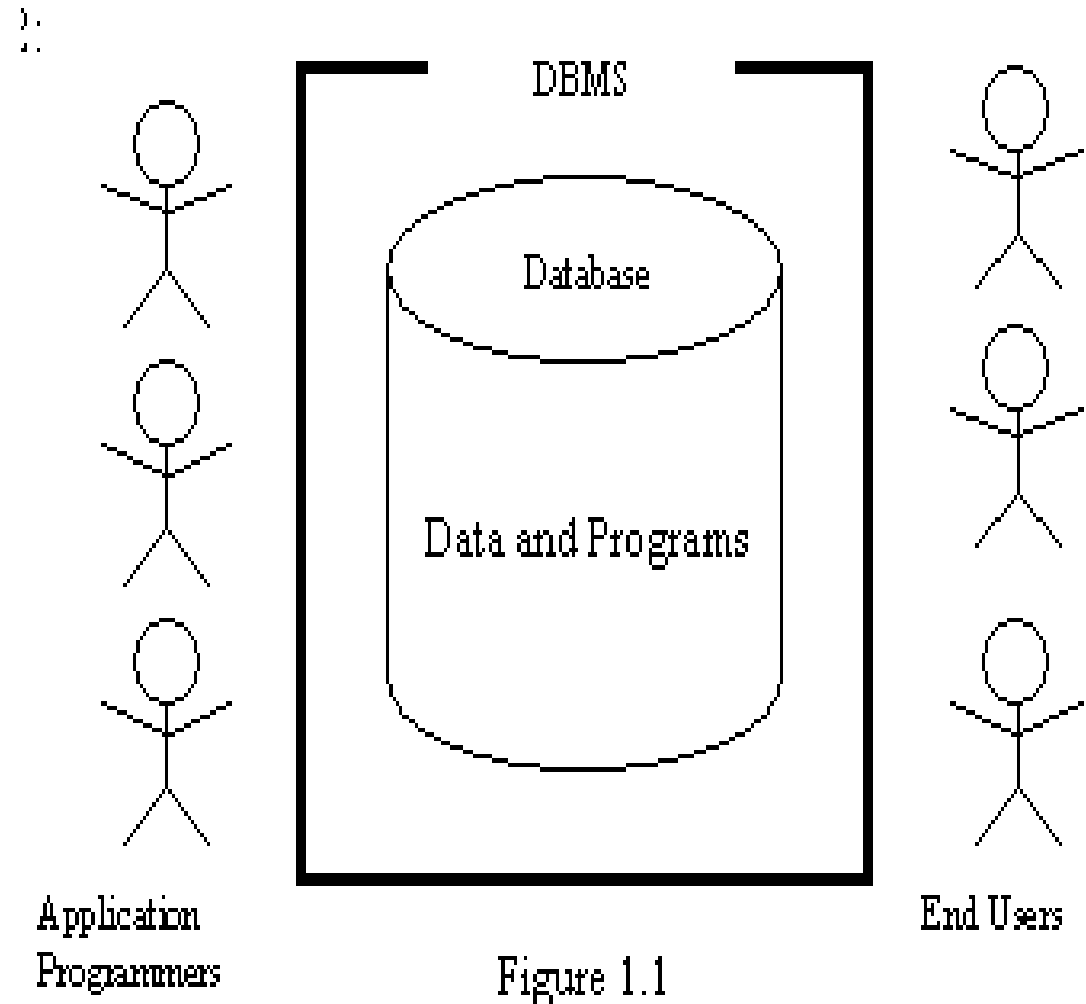


Figure 1.1

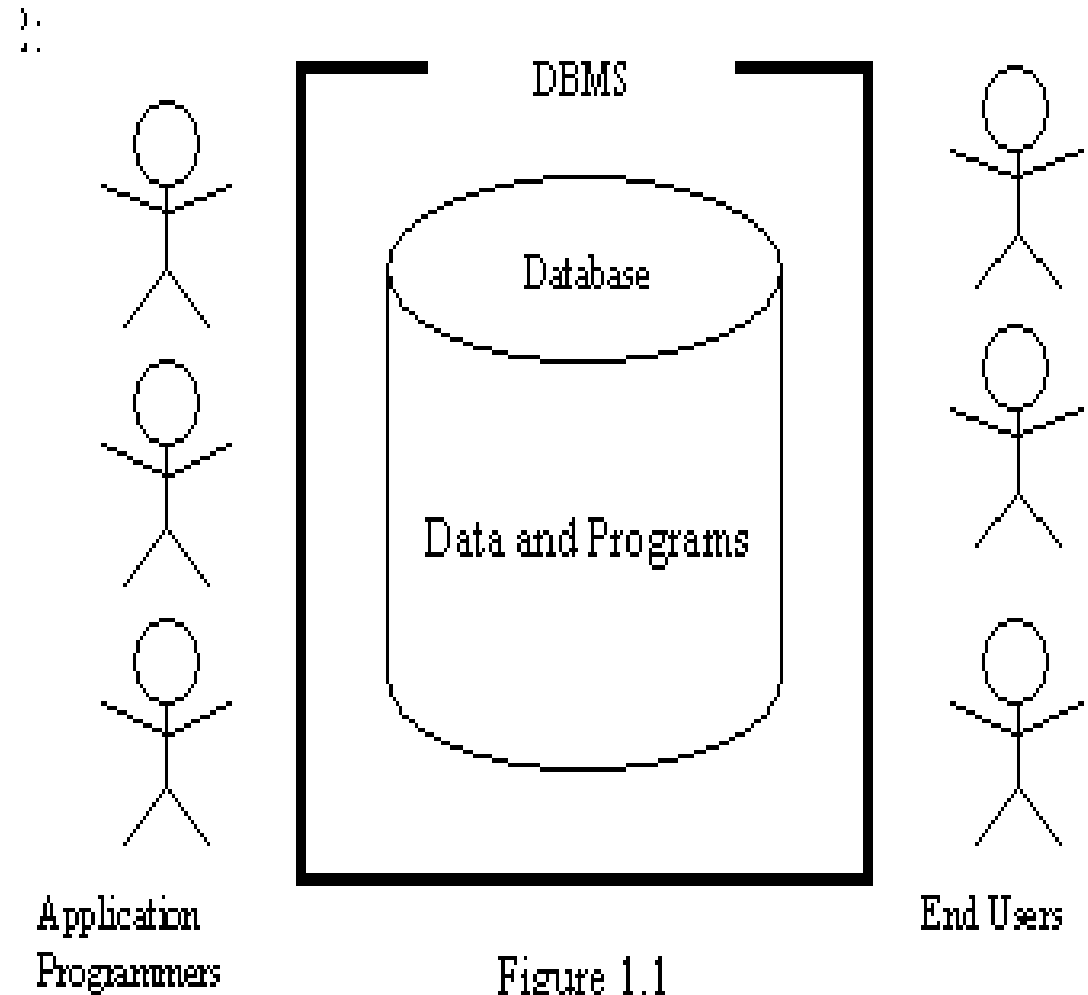
Components of DBMS: Hardware

The hardware is the actual computer system used for keeping and accessing the database. Conventional DBMS hardware consists of secondary storage devices, usually hard disks, on which the database physically resides.



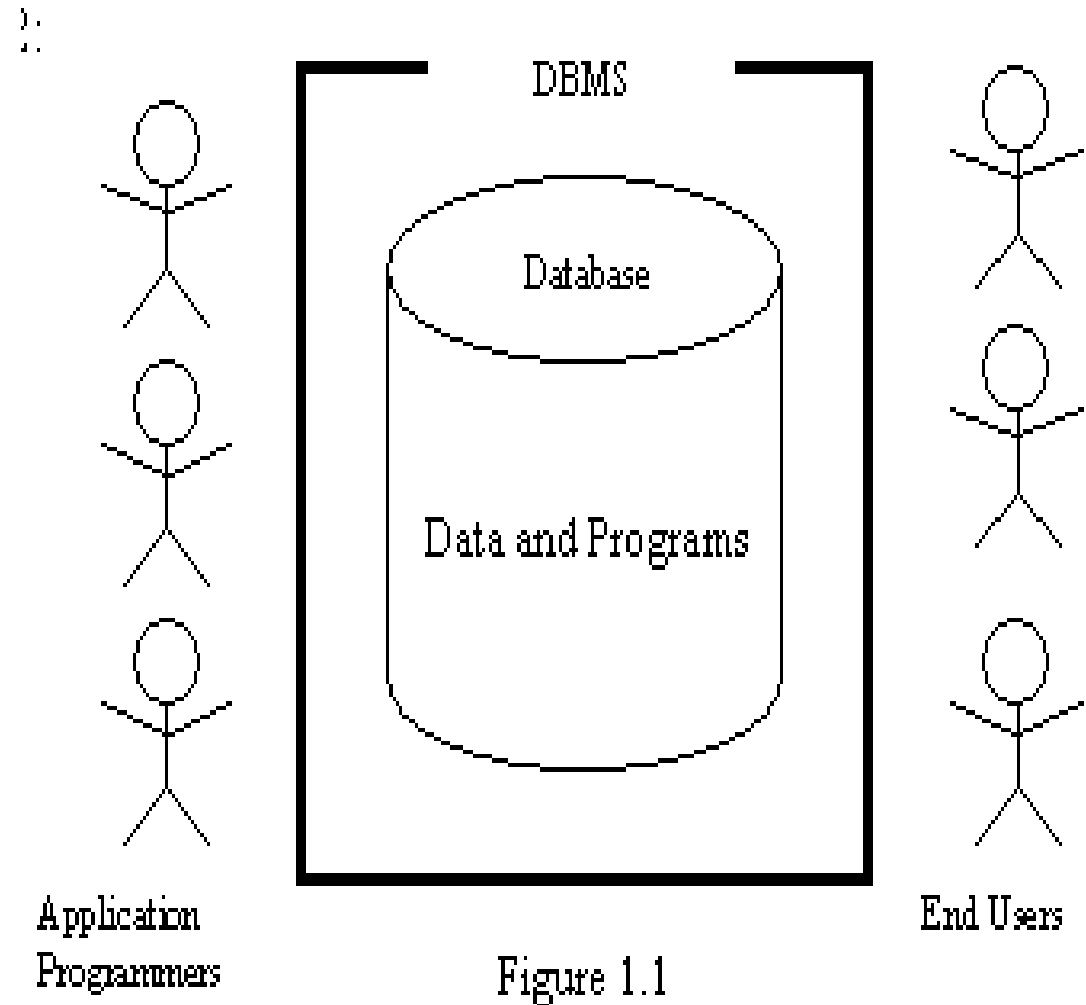
Components of DBMS: Software

The software is the actual DBMS. Between the physical database itself (i.e. the data as actually stored) and the users of the system is a layer of software, usually called the Database Management System or DBMS. All requests from users for access to the database are handled by the DBMS. One general function provided by the DBMS is thus the shielding of database users from complex hardware-level detail.



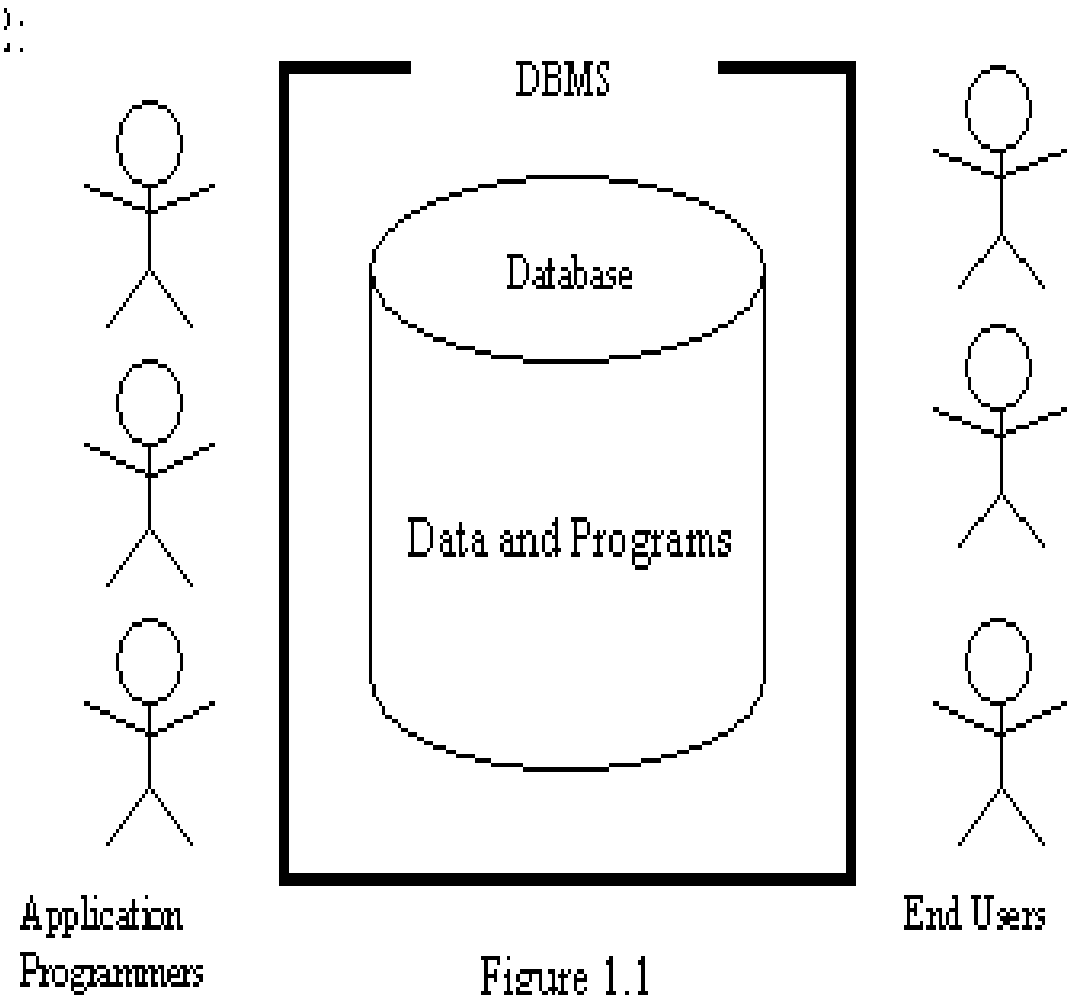
Components of DBMS: Data

- The database should contain all the data needed by the organization. One of the major features of databases is that the actual data are separated from the programs that use the data. A database should always be designed, built and populated for a particular audience and for a specific purpose.



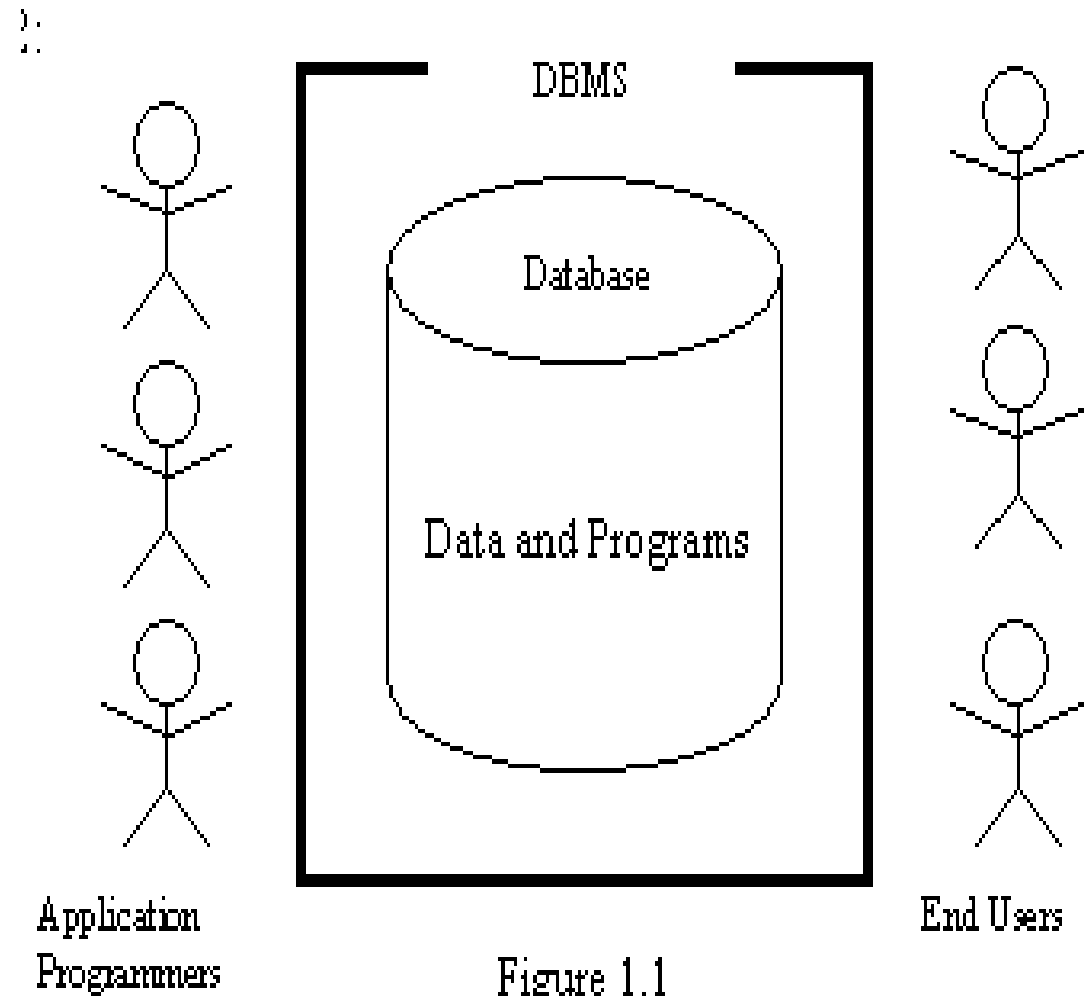
Components of DBMS: Users

- There are a number of users who can access or retrieve data on demand using the applications and interfaces provided by the DBMS.



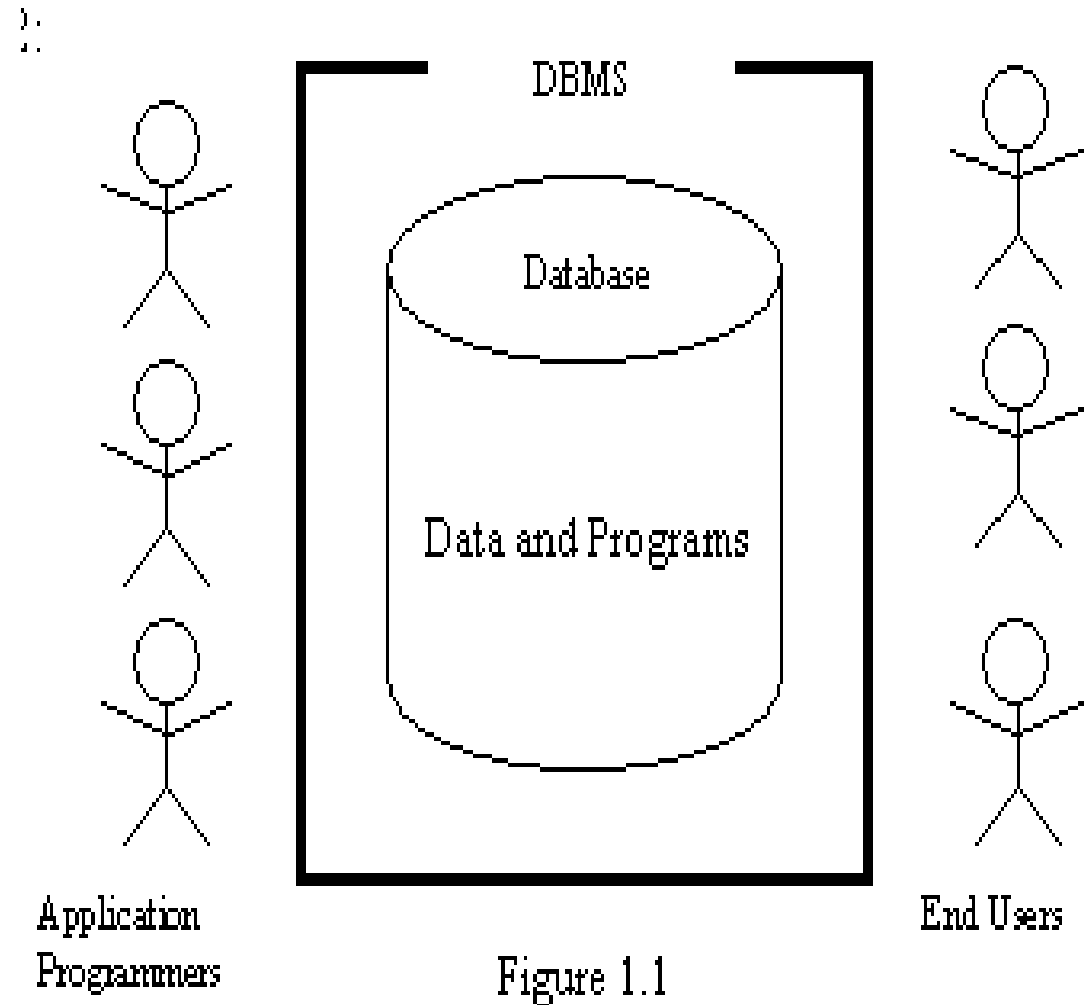
Components of DBMS: Users

- Naive Users
- Online Users
- Application Programmers
- Sophisticated Users
- DBA

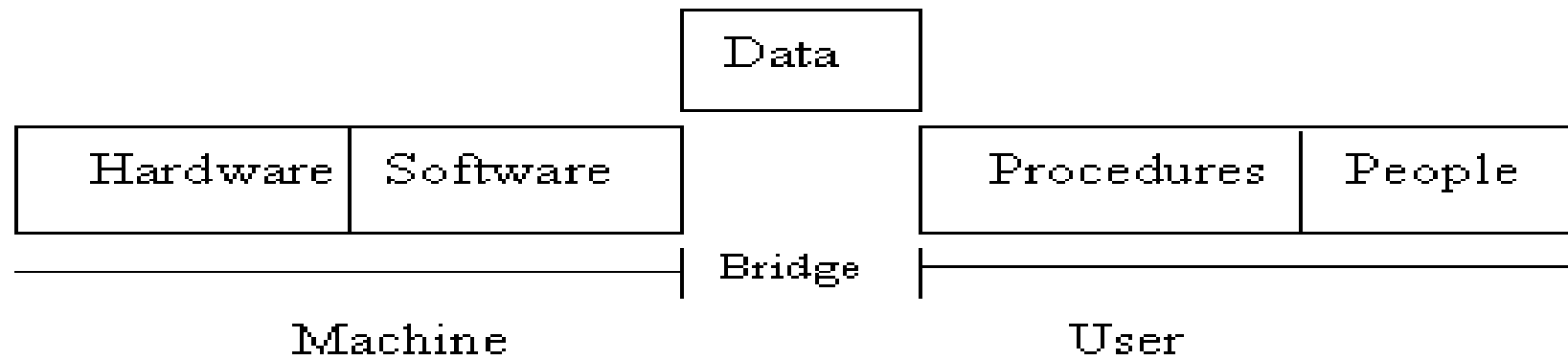


Components of DBMS: Procedures

- It defines the ways and means to access the data or information by the users.



Components of DBMS



Thanks
